

A Game Approach for Charging Station Placement Based on User Preferences and Crowdedness

Sangjun Bae¹, Inmo Jang, Sébastien Gros², Balázs Kulcsár³, and Jonas Hellgren

Abstract—The placement of electric vehicle charging stations (EVCSs), which encourages the rapid development of electric vehicles (EVs), should be considered from not only operational perspective such as minimizing installation costs, but also user perspective so that their strategic and competitive charging behaviors can be reflected. This paper proposes a methodological framework to consider crowdedness and individual preferences of electric vehicle users (EVUs) in the selection of locations for fast-charging stations. The electric vehicle charging station placement problem (EVCSP) is solved via a decentralized game theoretical decision-making algorithm and k -means clustering algorithm. The proposed algorithm, referred to as k -GRAPE, determines the locations of charging stations to maximize the sum of utilities of EVUs. In particular, we analytically present that 50% of suboptimality of the solution can be at least guaranteed, which is about 17% better than the existing game theoretical based framework. We show a few variants to describe the utility functions that may capture the difference in preferences of EVUs. Finally, we demonstrate the viability of the decision framework via three real-world data-based experiments. The results of the experiments, including a comparison with a baseline method are then discussed.

Index Terms—Electric vehicle charging station placement, game theory, clustering algorithm.

I. INTRODUCTION

ELECTRIC Vehicle Charging Stations (EVCSs) can be viewed as supporting components in the development of Electric Vehicles (EVs) and plug-in hybrid electric vehicles. As these vehicles are expected to become predominant in many EU countries in the near future [1]–[5], relevant studies facilitate their commercialization and industrialization. In particular, fast-charging EVCSs will play a critical role in the EV

industry. Nevertheless, EVCSs will have a limited number of charging slots, and limited total charging power dictated by the strength of their connection to the power grid and by their rated power. This necessitates a fair and effective distribution of the limited resources.

In recent years, considerable research attention has been devoted to fast EVCSs for urban design and planning, as they play a central role in all concepts of sustainable and smart cities such as impacts on the electrical grid and the environment [6]. Fast-charging EVCSs can have a charging speed 12 times higher than that of home-charging [7]. Furthermore, they positively influence on increasing the rate of penetration of EVs [8]. The requirements of drivers/users should be considered in the placement problems of EVCSs [8], [9]. Hence, EVCSs may reduce (among others) the social costs (e.g., travel distance, congestion by unbalanced distribution) generated by EV Users (EVUs), and may simultaneously increase the satisfaction level of EVUs.

In this regard, an urgent problem for smart infrastructure planners is the optimal placement of charging stations. This problem is often labeled as the EVCS Placement Problem (EVCSP) [10]. To solve the resulting EVCSP, in this paper, we propose to use k -GRAPE, i.e., a k -means clustering algorithm [11] with GGroup Agent Partitioning and Placing Event (GRAPE) [12] to maximize the sum of individual utilities of all the EVUs (named social cost). The method consists of two subsequent steps. In the first-step of the algorithm solves the decision-making problem of the EVUs modeled as *hedonic games* [13]. The objective of this game is to find a *Nash stable* partition that is agreed by all the EVUs [14]. In this view, we model a conflict situation where self-interested EVUs are willing to form groups to improve their own interest. Where each EVU consider the size of the group, called *crowdedness*, to which it belongs instead of the identities of the members (i.e., privacy-preserved information). We use the same assumptions for such games: each selfish EVU is willing to choose a unique EVCS by considering the population density at the EVCS, which is arguably anonymous public information. Furthermore, the EVCS selection is based on the individual preferences of EVUs such as brand and local attractions (e.g., shopping malls, public parks) as well as crowdedness. In the second-step, the objective is to find optimal locations of the EVCSs that minimize the social costs generated by EVUs given a Nash stable partition. This step minimizes the variance of the travel distance between the EVUs and the EVCS selected (i.e., maximizing the sum of utilities of EVUs). We show the existence and convergence of the proposed framework if EVUs have social inhibition

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(e.g., preference toward an EVCS with either uncrowded or a specific brand). Furthermore, we analyze the lower bound of outcome's suboptimality and show that 50% is at least guaranteed. Three numerical experiments are conducted based on real-world data from Gothenburg, Sweden.

The main contributions of this study to solve the charging station placement problem from the EVUs' perspective are summarized as follows:

A. Strategic Charging Behavior Using the Hedonic Game Model

The hedonic game models the formation of groups of multi-players when the decisions of self-interest EVUs are conflicting. The objective of each EVU who has individual preferences is to improve their own utility. Therefore, the hedonic game is suitable for modeling the charging station selection situation where EVUs greedily select charging stations based on their individual preferences and crowdedness. To the best of our knowledge, this is the first study that formulates the decision-making problem of EVUs for charging station placement problem in a selfish game-theoretical framework called GRAPE (see [12]). In addition, we propose a few forms of utility functions that model the strategic charging behavior of EVUs.

B. Individual-Level EVCS Placement Using k -GRAPE

In [7], the authors determine locations of EVCSs based on the decision of the group of EVUs who are in the predetermined districts (group-level), in which each location of EVCS is at the center of each district. We solve the EVCSPP in the individual-level, in which the decision of each EVU affects the placement of the EVCSs. Also the location of each EVCS is determined without any restrictions such as center of districts. For the restriction-free location we propose the k -GRAPE algorithm, which can also allow us to consider the strategic and competitive decision-making of EVUs depending on their preferences. We prove the existence, convergence and suboptimality of the proposed algorithm. In particular, we analytically present that 50% of suboptimality of the solution can be at least guaranteed, which is about 17% better than the existing game theory-based framework [7].

C. Case Study With Real-World Data

We illustrate the proposed algorithm via three experiments based on real-world data from Gothenburg, Sweden. The experimental results show that the proposed framework can efficiently solve the EVCSPP, including the preferences of EVUs. It is shown that the proposed approach outperforms the baseline approach [15], [16].

The rest of this paper is organized as follows. Section II discusses the related work. In Section III, we formulate an EVCSPP to maximize the total sum of utilities of EVUs. Section IV describes the k -GRAPE game-theoretical decision-making framework. Three experiments are described and analyzed in Section V. The experiments tackle three EVCSPP cases in Gothenburg. The concluding remarks are presented in Section VI.

II. LITERATURE REVIEW

Many studies have been conducted to determine the best locations of EVCSs [7]–[10], [15]–[31], [31]–[36]. The existing literature on the EVCSPP can be divided into two categories depending on the point of view adopted, namely 1) from the EVCS point of view; or 2) from the EVUs' point of view.

A. Perspective of EVCSs

Many previous studies have approached this problem from the point of view of the EVCSs, i.e., how to minimize the installation and/or operating costs and how to maximize the annual operating income [9], [15], [16], [23], [25], [26], [33], [36]. In [17], the authors proposed an optimization approach for the placement including its size to minimize the investment and operation cost of charging stations, and facilitate consumers to charge. The work in [9] proposed a cost model for the EVCSPP considering the investment, operation, and depreciation costs, and suggested EVCS layout planning by minimizing the cost. In [29], the authors attempted to minimize the total installation cost of new charging stations and existing ones. In this work, the network topology and origin-destination trips of EVUs in the network were considered. In [31], the authors proposed an approach for solving multi-criteria decision-making problem. The approach utilizes a grey decision-making trial and evaluation laboratory for determining criteria weights and uncertain linguistic multi-objective optimization by ratio analysis plus full multiplicative form to evaluate and select the optimal location of EVCS. In [25], the authors formulated the EVCS placement problem for electric buses in the public transportation system as an integer linear programming problem. They proposed an algorithm to solve the problem and showed effectiveness. In [31], the authors proposed an integrated multiple criteria decision making approach using a grey decision making trial and evaluation laboratory method to determine the most suitable EVCS site. In [33], the authors formulated a bi-level optimization problem to determine the charging station placement problem and the number of chargers at each charging station to maximizing profit. The problem considers network congestion, queuing, elastic demand and charging price elasticity. They solved the problem using the genetic algorithm in an iterative way. In [36], the authors proposed a framework to solve a multi-objective optimization for the charging station placement problem. The objective functions are minimizing the cost, guaranteeing grid stability and feasible charging station accessibility. The found results showed superior results compared to the benchmark algorithms with respect to the suggested measures.

B. Perspective of EVUs

Many studies [7], [10], [16], [18], [21], [22], [27], [30], [34], [35] have considered the perspective of EVUs (common preferences), which is a factor that influences the EVCSPP. In [16], the EVCSPP was solved by using a weighted Voronoi diagram to minimize the social cost incurred by the charging behavior of EVUs. The authors assumed that each EVU selects

the nearest EVCS. In [18], the EVCSPP was addressed to minimize waiting and travel time from the perspective of the cooperative EV model using genetic programming. They showed that the optimal solution for the EVCSPP is to achieve the minimum possible travel time. In [27], the authors tackled the EVCSPP in the public parking lots of Seattle. A mixed-integer programming problem to minimize the total EVCS access costs of EVUs has been proposed. In [10], the authors focused on human factors rather than technological factors. In [30], the authors addressed facility location allocation problems to minimize the transportation time and cost. They proposed a hybrid intelligent algorithm to solve the multi-objective optimization problem. However, there exist regional constraint issues. In [22], the authors proposed a multistage EVCSPP that predicts the aggregated charging demand of EVs, and balances the benefits of EVCSs. In [28], the authors developed bi-level model to optimize the EVCSPP with consideration of the EVUs' driving range. They showed the driving range has a large effect on the EVCSPP. Most recently, the authors in [7] proposed a game-theoretical framework to determine the locations of EVCSs such that each EVU strategically and competitively selects a charging station to minimize both the travel time and waiting time. The study also modified the multi-layer optimization problem into a single optimization problem, and performed practical experiments in Singapore. To the best of our knowledge, the study [7] is the only one that tackles the EVCSPP by considering the strategic behavior of EVUs using a game-theoretical approach. In [7], EVUs are both strategic and competitive in terms of travel and queuing time. Furthermore, in [7], the authors attempted to apply time-related factors accurately from the measured data. However, this resulted in constraining the number of EVCSs, computational issues when a large number of EVUs were introduced, and position restrictions (i.e., the solution was limited to zone-centered EVCS placement). In [34], the authors proposed two heuristic algorithms and five sub-modular objective functions for charging station placement problem. All the objective functions share customers' interests such as charging willingness, charging demand, distance, etc. The proposed algorithms were evaluated on the database in Beijing. In [35], the elastic charging demand with respect to both travel distance and waiting time was considered for the fast-charging station placement and sizing issue. One assumption applied in the paper is one charging station at the center of each predetermined zone, which is quite restricted.

In the future, in large-scale urban areas, the selection of dynamically changing charging stations of EVUs and their placement problems should be adequately addressed. In such a case, the selections of EVUs can be affected not only by intuitive factors (i.e., charging price, travel time, waiting time), but also by the preferences of individual drivers (i.e., brands, travel distance from local attractions). In this study, we address different EVU preferences and incorporate them into the EVCSPP. We assume that the charging habits of EVUs may be influenced by individual preferences such as fidelity to a brand, local social attractions as well as crowdedness and travel distance (see Figure 1).

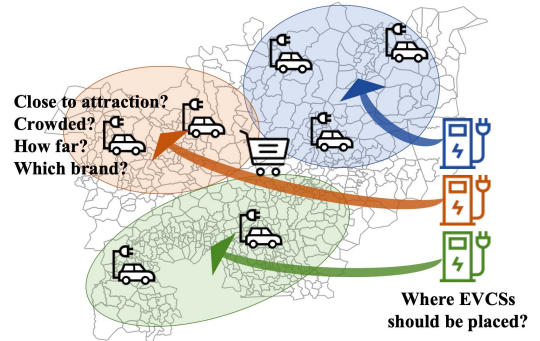


Fig. 1. Schematic depiction of considered electrical vehicle charging station placement problem.

III. PROBLEM FORMULATION

Adequate EVCS locations arguably enable EVUs to utilize the EVCS facilities efficiently in terms of their own utilities and from the operational perspective of EVCSs. Therefore, the objective of placement of the EVCSs is not only to balance the number of customers between EVCSs, but also to address the EVU satisfaction by providing those infrastructures. In this section, we first suggest charging cost models (utility functions) of the EVUs. Then, the location planning for EVCSs is approached from an EVU perspective, in other words, EVUs' individual preferences and crowdedness are considered. Their satisfaction can be considered by taking into account the strategic behavior to maximize their utilities. For better understanding of the definitions, we summarize the notations in Table I.

A. EVU Model: Utility Functions

Charging preferences vary from person to person. Some people might prefer to visit an EVCS nearby, whereas others might prefer to visit only a certain EVCS brand, or favor social attractions. Moreover, if EVUs prefer to spend the least amount of time at the EVCS, they may prefer to visit an EVCS having more available charging slots (capacity) than other EVCSs. In contrast to most existing studies, we consider the individual preferences of EVUs, such as brands, travel distance, and social attractions. We suggest four baseline utility functions depending on individual preferences. For the sake of consistency in the paper, in this section, we intentionally use $\mathbf{X}$ as an input argument in the utility terms although the decision variable of the EVUs is only $\mathbf{P}$, which is described in Section IV.

1) *Travel Distance*: EVUs favor visiting nearby EVCSs. Therefore, the travel distance, $d_i(\mathbf{P}, \mathbf{X})$, is one of the most important factors in EVCS selection. Furthermore, the travel distance for charging can be considered as social cost. It is common to utilize the travel distance for other factors such as fuel consumption, congestion level and travel time by multiplying the travel distance with the relevant weight [7], [37], [38]. We utilize the travel distance as a cost as is, i.e., the travel distance is proportional to the cost.

2) *Crowdedness*: We consider the number of expected EVUs at EVCSs as one of the individual preference factors.

TABLE I
DEFINITION OF VARIABLES AND ACRONYMS

Variable	Description
$\mathcal{A}$	a set of n_a EVUs
a_i	the i -th EVU
u_i	the individual utility of EVU a_i
$\mathcal{E}$	a set of n_e EVCSs
e_j	the j -th EVCS
r_j	the capacity at EVCS e_j
s_{ij}	a binary decision variable that indicates whether EVU a_i selects EVCS e_j
$\mathbf{X}$	a set of n_e EVCSs' locations
x_j	the j -th EVCS's location
g_j	an EVCS-specific group for e_j
$\mathcal{F}$	a set of n_f attractions
f_k	the k -th attraction
$\mathbf{P}$	a partition: a disjoint set that groups the EVU set $\mathcal{A}$, $\mathbf{P} = \{g_1, g_2, \dots, g_{n_e}\}$
$\mathbf{P}(i)$	index of the EVCS to which EVU a_i is assigned given $\mathbf{P}$
$C_c(k)$	the normalized closeness centrality for f_k
Acronym	Definition
EEDP	EVUs' EVCSs decision-making problem
ELUP	EVCS location update problem
EVCS	Electric vehicle charging station
EVCSPP	EVCS placement problem
EV	Electric vehicle
EVU	Electric vehicle user
SPAO	Single peaked at one

Due to the limited EVCS resources, the decision of each EVU might influence the decisions of others, mostly owing to the limited EVCSs resources, which must be shared between EVUs. For example, EVUs might prefer to visit an EVCS that has more available slots than the EVCSs that are closer but have fewer available slots with an assumption that booking systems are not considered at the EVCSs planning level. This can be interpreted in a few ways such as queuing time [7]. We define the “crowdedness” utility function for EVU a_i charging at EVCS e_j with the number of EVUs in group $|g_j|$ (including itself) as follows:

$$u_i^{crowd}(\mathbf{P}, \mathbf{X}) = \frac{r_j}{|g_{\mathbf{P}(i)}|}, \quad (1)$$

where r_j is the capacity at EVCS e_j , and $g_{\mathbf{P}(i)}$ is the group that EVU a_i selected at given $\mathbf{P}$. Note that the background of the word *crowdedness* is that, although we can count the number of EVUs at an EVCS in this study, it is difficult to count the number of EVUs in practice. Hence, we define “crowdedness” to indicate the ambiguity-included population density at EVCSs.

3) *EVCS Brand*: The brand preferences of each EVCS is one of the factors that determine the decision of EVUs. It is expected that EVUs will have EVCS brand preferences in a similar manner to a gas station market [39]. The relationship between brand preferences and purchase intent has been studied in [40]. For example, brand preferences can result from marketing strategies such as millage, service quality, and complementary services. Therefore, as shown in Figure 2,

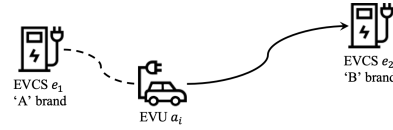


Fig. 2. Individual brand preferences based charging station selection. The EVU a_i decides to visit the EVCS e_2 (“B” brand) despite its significant travel distance.

the EVU a_i might visit the EVCS e_2 , which is far away but the preferred brand, rather than visiting the nearby EVCS e_1 . A weighting of the crowdedness utility function can be used to model individual preferences for a particular EVCS brand. Thus, the inclusion of brand preferences in the EVU utility can be given by

$$u_i^{brd}(\mathbf{P}, \mathbf{X}) = u_i^{crowd}(\mathbf{P}, \mathbf{X}) - w_{ij}^{brd} \cdot d_i(\mathbf{P}, \mathbf{X}), \quad (2)$$

where w_{ij}^{brd} indicates the brand preferences weight by which EVU a_i prefers EVCS e_j 's brand. Under the assumption that each EVCS has its own brand (i.e., n_e brands exist), we categorize the weight w_{ij}^{brd} of individual brand preferences into levels as follows:

$$w_{ij}^{brd} = \begin{cases} w^{1^{st}cat}, & \text{if } b_i^{brn(1)} = e_j \\ w^{2^{nd}cat}, & \text{else if } b_i^{brn(2)} = e_j \\ \vdots, & \vdots \\ w^{n_e-1^{th}cat}, & \text{else if } b_i^{brn(n_e-1)} = e_j \\ w^{n_e^{th}cat}, & \text{otherwise,} \end{cases} \quad \forall e_j \in \mathcal{E}, \quad \forall a_i \in \mathcal{A}, \quad (3)$$

where $b_i^{brn(j)}$ is the j -th preferred brand of EVU a_i . Normalized weights are being used, i.e., the weighting category $w^{j^{th}cat}$, $\forall e_j \in \mathcal{E}$ is a positive value between 0 and 1.

4) *Attractions*: EVUs should also consider charging in the middle of their journey because of various reasons such as low state of charge and traffic jam. Furthermore, they might prefer to charge around social attractions such as supermarkets, shopping malls, and coffee shops. Figure 3 shows a schematic example considering the effect of an attraction on the decision of EVUs. Figure 3(a) shows the result of selecting an EVCS between two EVCSs based on the travel distance. Here, 10 EVUs select EVCS e_A , and the other 10 EVUs select EVCS e_B . Figure 3(b), which is the case where there is a shopping mall f_S near EVCS e_A , shows that four EVUs (with right curved arrow) change their decision from EVCS e_B to EVCS e_A because they prefer to visit the EVCS near the shopping mall. Particularly, if the shopping mall is highly preferred by the EVUs, it might be possible to select an EVCS near the shopping mall although it is far away, such as the EVU at the bottom left in this case.

A weight can represent how close the attractions to the EVCSs, which affects the decisions of the EVUs. These attraction-related individual preferences can be interpreted as follows:

$$u_i^{att}(\mathbf{P}, \mathbf{X}) = w_{ijk}^{att} \cdot u_i^{crowd}(\mathbf{P}, \mathbf{X}), \quad (4)$$

where weight w_{ijk}^{att} is a normalized travel distance value (0~1] between EVU a_i and attraction f_k . Here, we only consider the

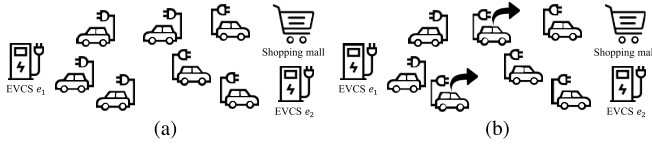


Fig. 3. The figures show the influence of the attractor (shopping mall). The direction of each EVU represents its EVCS selection. (a) EVCS selection only based on travel distance. (b) EVCS selection based on both travel distance and individual preferences on the shopping mall, and curved arrows indicate changed decisions of the EVUs from the EVCS e_1 to the EVCS e_2 due to the shopping mall.

closest attraction point f_k from EVCS e_j for the utility term $u_i^{att}(\mathbf{P}, \mathbf{X})$. Weight w_{ijk}^{att} can be obtained as follows:

$$w_{ijk}^{att} = \begin{cases} \text{dn}(a_i, f_1), & \text{if } \min_{f_k \in \mathcal{F}} (||x_j, f_k||) = f_1 \\ \text{dn}(a_i, f_2), & \text{else if } \min_{f_k \in \mathcal{F}} (||x_j, f_k||) = f_2 \\ \vdots, & \vdots \\ \text{dn}(a_i, f_{n_f-1}), & \text{else if } \min_{f_k \in \mathcal{F}} (||x_j, f_k||) = f_{n_f-1} \\ \text{dn}(a_i, f_{n_f}), & \text{otherwise,} \end{cases} \quad \forall e_j \in \mathcal{E}, \quad \forall a_i \in \mathcal{A}, \quad \forall f_k \in \mathcal{F}, \quad (5)$$

where $\text{dn}(a_i, f_k)$ is the normalized travel distance between EVU a_i and attraction f_k . Therefore, the weight w_{ijk}^{att} indicates the preferences of EVU a_i with respect to attraction f_k that is the closest attraction to EVCS e_j .

The behavior of each EVU is to pursue its highest utility greedily and unilaterally. We adopt the condition, called SPAO, that enables the proposed approach to satisfy the behavior.

Definition 1 (SPAO Condition): Given $\mathbf{X}$ and $\mathbf{P}$, for EVU a_i , let $\tilde{\mathbf{P}}_{-i} = \cup_{a_l \in \mathcal{A} \setminus \{a_i\}} \tilde{\mathbf{P}}_l$ denote all the possible deviated partitions by any of all the other EVUs. The individual utility of EVU a_i is SPAO (Single-Peaked-At-One) if it holds for any $\mathbf{X}$ and $\mathbf{P}$ that $u_i(\mathbf{P}, \mathbf{X}) \geq u_i(\mathbf{P}', \mathbf{X})$ for $\forall \mathbf{P}' \in \tilde{\mathbf{P}}_{-i}$ such that $|g_{\mathbf{P}(i)}| < |g_{\mathbf{P}'(i)}|$.

In other words, the condition can be interpreted as follows: the utility of each EVU is monotonically decreased whenever a new EVU joins the same group to which the former belonged, regardless of the locations and partition of the current EVCS. This indicates that EVUs satisfying this condition tend to avoid crowdedness.

EVUs make decisions based on various criteria. In Section V, we suggest multi-criteria utility functions by combining the baseline utility functions. Note that, in this study, we do not address how to determine the weights for multi-criteria utility functions. Further, other forms of social attractions can be added (e.g., schools, train stations).

B. EVCS Placement Problem

Let us introduce the EVCS placement problem considered in this paper and assumptions applied in this study.

Problem 1 (EVCSP): Given a set of n_e EVCSs $\mathcal{E} = \{e_1, e_2, \dots, e_{n_e}\}$, their locations $\mathbf{X} = \{x_1, x_2, \dots, x_{n_e}\}$ are to be determined. Each EVU in a set of n_a EVUs $\mathcal{A} = \{a_1, a_2, \dots, a_{n_a}\}$ selects one EVCS $e_j \in \mathcal{E}$ for charging. EVUs that select the EVCS $e_j \in \mathcal{E}$ for charging are in group

$g_j \in \{1, 2, \dots, n_a\}$ where the set of groups is defined as the partition $\mathbf{P} = \{g_1, g_2, \dots, g_{n_e}\}$. The charging station selection of each EVU is based on the EVU a_i utility $u_i : \mathbf{P} \times \mathbf{X} \mapsto \mathbb{R}$, which is a function of the location of EVCS x_j where the EVU is going for charging itself, EVU individual preferences, and the expected number (including itself) of EVUs for the EVCS (i.e., $|g_j|$). The objective of the EVCSP is to determine a set of n_e EVCSs locations $\mathbf{X} = \{x_1, x_2, \dots, x_{n_e}\}$ so as to maximize the sum of individual utilities of all the EVUs based on their self-interested behavior. The problem can be defined as follows:

$$\max_{\{\mathbf{X}, s_{ij}\}} \sum_{\forall a_i \in \mathcal{A}} \sum_{\forall e_j \in \mathcal{E}} u_i(\mathbf{P}, \mathbf{X}) s_{ij} \quad (6)$$

$$\text{subject to } \sum_{\forall e_j \in \mathcal{E}} s_{ij} = 1, \quad \forall a_i \in \mathcal{A} \quad (7)$$

$$s_{ij} = \begin{cases} 1, & \text{if } a_i \in g_j \\ 0 & \text{otherwise,} \end{cases} \quad \forall e_j \in \mathcal{E}, \quad \forall a_i \in \mathcal{A}, \quad (8)$$

$$g_j \cap g_{j'} = \emptyset, \quad \forall g_j, g_{j'} \in \mathbf{P}, j \neq j' \quad (9)$$

$$x_j \neq x_{j'}, \quad \forall x_j, x_{j'} \in \mathbf{X}, j \neq j' \quad (10)$$

where s_{ij} is a binary decision variable that indicates whether EVU a_i selects EVCS e_j . Note that we can simply derive s_{ij} for Eq.(6) from $\mathbf{P}$ and vice versa. The utility of i -th EVU, u_i , includes individual preferences determining their strategic behaviors, more details of the utility function are described in Section III-A. one EVCS for charging and EVCSs cannot be located at the same position. Note that, we use the partition $\mathbf{P}$ instead of $|g_j|$ for consistency with Section IV.

Assumption 1 (Heterogeneous EVUs With the Same Charging Time and Price): We consider a large-scale system composed of the decisions of heterogeneous EVUs and EVCS placement. For simplicity, we assume that the EVUs might have different individual preferences but they spend the same amount of charging time at all the EVCSs. Also, electricity prices are the same in different EVCSs.

Assumption 2 (Single EVCS Selection): For simplicity, we assume that each EVU selects only one EVCS to charge although there could be other strategic ways to charge such as visiting multiple EVCSs, charging only enough to go home.

Assumption 3 (Privacy Preserved Data-Based Preferences of Each EVU): EVU a_i only knows its own preferences and individual utility u_i with regard to every EVCS, while not being aware of those of the other EVUs. However, EVUs can recognize how many other EVUs currently choose a specific EVCS by communicating with the EVCSs only. Hence, the individual EVU information is not public, but the privacy preserved data of anonymous EVUs (e.g., the expected number of EVUs at each EVCS) is public.

Assumption 4 (No Geographical Limits on Location of EVCS): In order to enlarge the solution space, we assume that EVCSs will be located within a planning region without any geographical restrictions. This assumption is valid in well-planned and high-density road network urban areas where EVCSs can be installed almost anywhere.

IV. SOLUTION TO THE GAME-THEORETIC EVCSPP

This section describes our proposed approach to address the EVCSPP. Although the objective function Eq.(6) is to maximize the aggregate utility of individual EVUs, it would not be agreeable to them, even if it is achieved (i.e., EVUs will not cooperate with each other). In the case of orchestrated (cooperative) approaches such as negotiation and voting model, some EVUs may have to be individually worse off to achieve the global optimum. In this context, rather than pursuing the global optimum, it is better to make the EVUs satisfied with a certain outcome. Consequently, each EVU will select an EVCS such that its own utility is maximized. For this, we apply a game-theoretical framework for EVCSPP. In game theory, a Nash equilibrium (or Nash stable partition) is a status where no EVU has any gain by unilaterally changing its own strategy.

In this study, we address the EVCSPP by splitting it into two sub-problems: (1) given EVCS locations $\mathbf{X}$, EVUs' EVCSs Decision-making Problem (EEDP), which can be modeled as a game-theoretic problem; (2) given partition $\mathbf{P}$, the EVCS Location Update Problem (ELUP).

Problem 2 (EEDP): At a given set of EVCSs' locations $\mathbf{X}$, there exist a set of EVUs $\mathcal{A} = \{a_1, a_2, \dots, a_{n_a}\}$ and a set of EVCSs $\mathcal{E} = \{e_1, e_2, \dots, e_{n_e}\}$. Each EVU can only select one EVCS for charging based on its individual preferences $u_i(\mathbf{P}, \mathbf{X})$. The objective of EEDP is to find an assignment that maximizes the sum of the individual utilities of all the EVUs. The problem described above is defined as follows:

$$\max_{\{\mathbf{P}\}} \sum_{\forall a_i \in \mathcal{A}} u_i(\mathbf{P}, \mathbf{X}) s_{ij}, \quad (11)$$

subject to Eq.(7) and Eq.(8). Using $\mathbf{P}$ as a decision variable for solving Eq.(11) is the same as using s_{ij} to find a Nash stable partition. Note that a solution of Eq.(11) is Nash stable partition $\mathbf{P}^*$.

Problem 3 (ELUP): Given a $\mathbf{P}$ that contains the decision of each EVU, the objective of ELUP is to update a set of locations of n_e EVCSs $\mathbf{X} = \{x_1, x_2, \dots, x_{n_e}\}$ to maximize the sum of the individual utilities of all the EVUs. The problem described above is defined as follows:

$$\max_{\{\mathbf{X}\}} \sum_{\forall a_i \in \mathcal{A}} u_i(\mathbf{P}, \mathbf{X}) s_{ij}. \quad (12)$$

Note that, Eq.(12) is valid when the utility function u_i contains the travel distance as one of the factors (i.e., Eq.(1) and Eq.(4) are not applicable here). We propose an algorithm based on game theory to deal with Problem 2 to obtain the selfish strategies of EVUs given the locations of EVCSs, whereas the approach for Problem 3 updates the locations to minimize the global utility of the EVUs. Figure 4 shows the flow of the proposed approach.

A. Step1: Clustering EVUs

Given the information of EVCSs (e.g., brand, capacity, location), each EVU will attempt to maximize its self-interested utility, not necessarily considering the utilities of other EVUs. Competitive and strategic behaviors can be modeled from

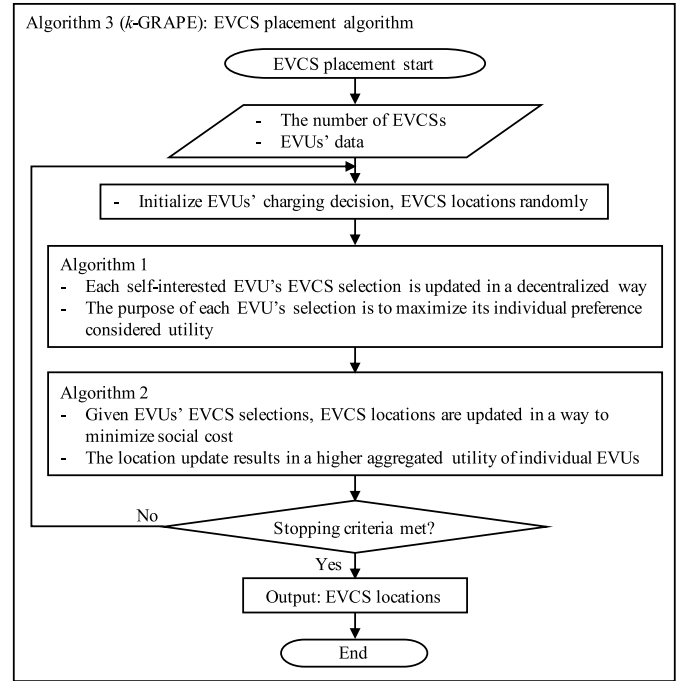


Fig. 4. Flowchart for the proposed approach (Algorithm 3).

game-theoretic perspectives. Here, it is assumed that each EVU can observe the tentative information in terms of how many other EVUs are willing to go to each EVCS (Assumption 3). If all of them are satisfied with their decisions with regard to the preferred EVCSs, the EVUs can be mutually and exclusively grouped for each EVCS. Otherwise, some of the EVUs that are not satisfied with the current status will deviate from it, rather preferring an alternative EVCS. This agreeable status can be described as the concept called *Nash stable partition*, which is formally defined as follows.

Definition 2 (Deviated Partition): An instance of EVUs' EVCSs Decision-making Problem (EEDP) is a tuple $(\mathcal{A}, \mathcal{E}, \mathcal{P})$ that consists of (1) $\mathcal{A} = \{a_1, a_2, \dots, a_{n_a}\}$, a set of n_a EVUs; (2) $\mathcal{E} = \{e_1, e_2, \dots, e_{n_e}\}$, a set of n_e EVCSs; and (3) $\mathcal{P} = \{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_{n_a}\}$, an n_a -tuple of preference relation of the EVUs. Given an instance $(\mathcal{A}, \mathcal{E}, \mathcal{P})$ of EEDP, EVU a_i may unilaterally deviate to group g_j as the user prefers EVCS e_j in the context. We refer to the resultant partition as a *deviated partition*, denoted by $\mathbf{P}_{i \rightarrow j}$.

For example, given $\mathbf{P} = \{\{a_1, a_2\}, \{a_3\}, \{a_4\}\}$, $\mathbf{P}_{1 \rightarrow 2}$ will be $\{\{a_2\}, \{a_3, a_1\}, \{a_4\}\}$. Note that $\mathbf{P}_{i \rightarrow \mathbf{P}(i)} = \mathbf{P}$. The set of all possible deviated partitions by EVU a_i is denoted by $\tilde{\mathbf{P}}_i$.

Definition 3 (Nash Stable Partition): Given $\mathbf{X}$, a partition $\mathbf{P}$ is *Nash stable* if, for every EVU $a_i \in \mathcal{A}$, it holds that $u_i(\mathbf{P}, \mathbf{X}) \geq u_i(\mathbf{P}_{i \rightarrow j}, \mathbf{X}), \forall e_j \in \mathcal{E}$.

Given an instance of Problem 2, the proposed algorithm to find a Nash stable partition is presented in Algorithm 1. Here, the superscripts indicate the local information known to EVUs.

Each EVU a_i has local variables for its decision such as $\mathbf{P}^i$, r^i , s^i , and *satisfied* ^{i} (line 2). Here, $\mathbf{P}^i$ is the locally known partition of the EVU; $r^i \in \mathbb{Z}^+$ is a positive integer variable to represent how many times $\mathbf{P}^i$ has evolved; $s^i \in [0, 1]$ is a uniform random variable generated whenever $\mathbf{P}^i$ is updated;

Algorithm 1 Decision-Making of Each EVU a_i

```

1: function EEDP( $\mathbf{P}$ ,  $\mathbf{X}$ )
  // Initialization
2:  $\mathbf{P}^i \leftarrow \mathbf{P}$ ;  $r^i \leftarrow 0$ ;  $s^i \leftarrow 0$ ;  $\text{satisfied}^i \leftarrow \mathbf{0}^{n_a \times 1}$ 
3: repeat
  // Make a new decision if necessary
4: if  $\text{satisfied}^i[i] = 0$  then
5:    $\hat{\mathbf{P}}^i \leftarrow \arg \max_{\mathbf{P}^i \in \tilde{\mathcal{P}}_i} u_i(\mathbf{X}, \mathbf{P}^i_{i \rightarrow j})$ 
6:   if  $\hat{\mathbf{P}}^i \neq \mathbf{P}^i$  then
7:      $\mathbf{P}^i \leftarrow \hat{\mathbf{P}}^i$ 
8:      $r^i \leftarrow r^i + 1$ ;  $s^i \in \text{unif}[0, 1]$ 
9:      $\text{satisfied}^i \leftarrow \mathbf{0}^{n_a \times 1}$ 
10:   end if
11:    $\text{satisfied}^i[i] \leftarrow 1$ 
12: end if
  // Distributed Mutual Exclusion
13: Broadcast  $M^i = \{r^i, s^i, \mathbf{P}^i, \text{satisfied}^i\}$ 
14: to the selected EVCS and
15: receive  $M^k$  from EVCSs  $\forall e_j \in \mathcal{E}$ 
16: Collect all the messages  $\mathcal{M}_{rcv}^i = \{M^i, \forall M^k\}$ 
17: for each message  $M^k \in \mathcal{M}_{rcv}^i$  do
18:   if  $(r^k > r^i)$  or  $(r^k = r^i \ \& \ s^k > s^i)$  then
19:      $M^i \leftarrow M^k$ 
20:   end if
21: end for
22: until  $\text{satisfied}^i = \mathbf{1}^{n_a \times 1}$ 
23: return  $\mathbf{P}^i$ 
24: end function

```

and satisfied^i is a binary array that includes the information about whether each EVU is satisfied with $\mathbf{P}^i$ such that it does not want to deviate from its current group. Given $\mathbf{P}^i$, EVU a_i examines which group is the most preferred among the other groups, assuming that the other EVUs remain at the current groups (line 5). Then the EVU selects the newly found group if it is more preferred than the current group. In this case, the EVU updates $\mathbf{P}^i$ to reflect its new decision, increases r^i , and generates a random time stamp s^i (line 6-10). In any case, the EVU has ascertained that the current group is the most preferred one, and hence, it is satisfied with $\mathbf{P}^i$ (line 11). Then, the EVU a_i sends its status to the selected EVCS via a message $M^i := \{r^i, s^i, \mathbf{P}^i, \text{satisfied}^i\}$, and receives M^j from all EVCSs (line 15).

As every EVU updates its locally known partition obtained from the EVCSs simultaneously, one of the partitions should be considered as if it were the partition updated by a deciding EVU at the previous iteration (refer to this partition as *the valid partition*). The distributed mutual exclusion part of Algorithm 1 enables the EVUs to identify the valid partition. Each EVU a_i collects all the messages from all EVCSs $\mathcal{M}_{rcv}^i$ (line 16). Based on the message set, the EVU identifies whether its partition $\mathbf{P}^i$ is valid. If there are partitions $\mathbf{P}^k$ such that $r^i < r^k$, then the EVU judges that partition $\mathbf{P}^k$ is better than $\mathbf{P}^i$. When $r^i = r^k$, higher time stamps s^i and s^k determine the valid partition. Now, as $\mathbf{P}^k$ is judged to be more valid, EVU a_i needs to re-examine whether there is a more preferred group

Algorithm 2 Location Update for Each EVCS e_j

```

1: function ELUP( $\mathbf{P}$ ,  $\mathbf{X}$ )
  // Initialization
2:  $\mathbf{P} \leftarrow \mathbf{P}^*$ 
  // Compute new location for EVCS  $e_j$ 
3: for  $j = 1$  to  $n_e$  do
4:    $x_j \leftarrow \frac{1}{|g_j|} \sum_{a_i \in g_j} a_i$ 
5: end for
6: return  $\mathbf{X}$ 
7: end function

```

in the next iteration (line 17-21). Depending on satisfied^i , each EVU undertakes the decision-making process again until $\text{satisfied}^i = \mathbf{1}^{n_a \times 1}$.

Lemma 1: Given $\mathbf{X}$, if an instance $(\mathcal{A}, \mathcal{E}, \mathcal{P})$ in which the current partition is Nash stable holds SPAO, the new instance $(\hat{\mathcal{A}}, \mathcal{E}, \hat{\mathcal{P}})$ having a new EVU $a_q \notin \mathcal{A}$ holding SPAO with regard to every EVCS in $\mathcal{E}$ also (1) satisfies SPAO; (2) contains a Nash stable partition; and (3) the maximum number of iterations required to converge to a Nash stable partition is $|\hat{\mathcal{A}}|$.

The proof of Lemma 1 is given in Appendix VI-A.

Theorem 1 (Existence of Algorithm 1): Given $\mathbf{X}$, if an instance $(\mathcal{A}, \mathcal{E}, \mathcal{P})$ holds the SPAO condition, then a Nash stable partition always exists.

The proof of Theorem 1 is given in Appendix VI-B.

Theorem 2 (Convergence of Algorithm 1): Given $\mathbf{X}$, if an instance $(\mathcal{A}, \mathcal{E}, \mathcal{P})$ holds the SPAO condition, then the number of iterations required to determine a Nash stable partition is at most $|\mathcal{A}| \cdot (|\mathcal{A}| + 1)/2$.

The proof of Theorem 2 is given in Appendix VI-C.

B. Step 2: EVCS Location Update

Given a partition $\mathbf{P}$ that is Nash stable, the location of each EVCS e_j can be updated to obtain a higher aggregated utility of individual EVUs. This is because the travel distance is the only decision variable that allows to maximize the aggregate utility given a partition. Therefore, given the partition the objective function Eq.(12) in Problem 3 is the Euclidean distance between EVU a_i in group g_j and the corresponding EVCS e_j .

In Algorithm 2, given the Nash stable partition $\mathbf{P}$ (i.e., all the EVUs have been assigned to EVCSs), EVCSs locations are updated by taking the average of all the locations of EVUs in that EVCS. This recalculation of EVCS locations results in a higher aggregated utility of individual EVUs.

C. k-GRAPE Algorithm for EVCS Placement Problem

Algorithm 3 first initializes random partition $\mathbf{P}$ and EVCS locations $\mathbf{X}$, and memory $\mathbf{D}$ (line 3). Each self-interested EVU a_i makes its own decision regarding EVCS e_j (line 6-7). Once all the EVUs have made their decision regarding EVCSs (i.e., a Nash stable partition $\mathbf{P}^i$), each EVCS location is re-located at the center of the average of the positions of EVUs, in which the EVUs select the EVCS for charging (line 8-9). This

Algorithm 3 k -GRAPE Algorithm

```

// Initialization
1: Initialize
2:  $t = 0$ 
3:  $\mathbf{P}, \mathbf{X}$  and memory  $\mathbf{D}$ ; Let  $\mathbf{P}^t = \mathbf{P}$  and  $\mathbf{X}^t = \mathbf{X}$ 
4: repeat
5:    $t = t + 1$ 
   // Update EVUs' (tentative) decisions
6:    $\mathbf{P}^{t-1} \leftarrow \mathbf{P}^t$ 
7:    $\mathbf{P}^t \leftarrow \text{EEDP}(\mathbf{P}, \mathbf{X}^t)$ 
   // Update EVCS locations
8:    $\mathbf{X}^{t-1} \leftarrow \mathbf{X}^t$ 
9:    $\mathbf{X}^t \leftarrow \text{ELUP}(\mathbf{P}^t, \mathbf{X}^{t-1})$ 
10:  Store  $(\mathbf{P}^t, \sum u_i(\mathbf{P}^t, \mathbf{X}^t))$  in  $\mathbf{D}$ 
11: until  $\mathbf{P}^t = \mathbf{P}^{t'}, \forall t' \in \mathbf{D}$ 
   // Find maximum utility sum in a recursive loop
12:  $\max_{(\mathbf{P}^t, \mathbf{X}^t)} \sum_{\forall t' \in \mathbf{D}'} u_i(\mathbf{P}^{t'}, \mathbf{X}^{t'})$ 
13: return  $\mathbf{P}^* \leftarrow \mathbf{P}^t, \mathbf{X}^* \leftarrow \mathbf{X}^t$ 

```

recalculation of EVCS locations results in a better EVCS location set to maximize the sum of individual EVU utilities. The process is continued until there is the same $\mathbf{P}^{t'}$ in memory $\mathbf{D}$ (line 11). It means that the same utilities are achieved for the EVUs and the same partitions will be repeated in following iterations. We define $\mathbf{D}'$ as follows

$$\mathbf{D}' = \{(\mathbf{P}^{t'}, \sum u_i(\mathbf{P}^{t'}, \mathbf{X}^{t'})), (\mathbf{P}^{t'+1}, \sum u_i(\mathbf{P}^{t'+1}, \mathbf{X}^{t'+1})), \dots, (\mathbf{P}^t, \sum u_i(\mathbf{P}^t, \mathbf{X}^t))\}.$$

Then, the algorithm returns $\mathbf{P}^*$, and $\mathbf{X}^*$ that maximize the sum of individual utilities (line 12).

Definition 4 (DIMC Utility Function): Given $\mathbf{X}$ and $\mathbf{P}$, the individual utility $u_i(\mathbf{P}, \mathbf{X})$ of the EVU is a *DIMC (Distance Included Multi-Criteria)* utility function if it holds both a reward that is not a function of the EVCS location, and a cost that is a function of the EVCS location.

Theorem 3 (Convergence of Algorithm 3): Given $\mathbf{X}$ and $\mathbf{P}$ that hold Definition 1 and Definition 4, the k -GRAPE algorithm converges to a local optimal solution of Problem 1 in a finite number of iterations.

The proof of Theorem 3 is given in Appendix VI-D.

Note that Algorithm 3 can be considered as a multi-leader-followers Stackelberg game in a sense that all the EVUs select their decisions first then the fast-EVCSs choose their decisions based on them, as similar to a paper [41].

D. Complexity of k -GRAPE Algorithm (Scalability)

We discuss the running time for the proposed framework to find a solution. The k -GRAPE algorithm has a time complexity of $O(T n_e n_a)$, where T is the number of iterations needed to complete the charging station selection process, n_e is the number of EVCSs, and n_a is the number of EVUs. For T , we refer to a unit time required for each EVU to proceed the main loop of Algorithm 1 (Line 4-23) as a time step. This iteration is depending on the communication network considered, which is assumed as fully-connected one in this

application (i.e., no extra iteration due to the communication overhead). Thanks to Theorem 2, the total required time steps until finding a Nash stable partition is $O(n_a^2)$. It is also obvious that the number of EVCSs is much smaller than the number of EVUs. Therefore, the time complexity becomes $O(n_a^3)$.

The number of required iterations in practice may less than $O(n_a^3)$. Firstly, the running time can be much less than the bound shown in Theorem 2. Because the theorem is based on the proof that a new EVU can be involved after a Nash stable partition of existing EVUs is found. Since Algorithm 1 do not need to find every Nash stable partition for each subset of the EVUs, unnecessary iterations can be reduced. Secondly, we may expect the running time reduction from Algorithm 3 (Line 4) as the initial partition at iteration t is the Nash stable partition at iteration $t - 1$.

E. Suboptimality Analysis of k -GRAPE Algorithm

This section investigates the *suboptimality lower bound* of Algorithm 3 in terms of the global utility (i.e., Eq.(6)). Given a partition $\mathbf{P}$ for any set of EVCS locations $\mathbf{X}$, the global utility value can be equivalently redefined as

$$J = \sum_{\forall a_i \in \mathcal{A}} u_i(e_{\mathbf{P}(i)}, |g_{\mathbf{P}(i)}|, \mathbf{X}) \quad (13)$$

where, for the convenient, we redefine $(\mathbf{P}, \mathbf{X})$ as $(e_j, |g_j|, \mathbf{X})$, which is a EVCS-group pair in a given set of EVCS locations (i.e., to charge at e_j with $|g_j|$ participants). Let $J_{k\text{-GRAPE}}$ and J_{OPT} represent the global utility of a Nash stable partition in any set of EVCS locations by the proposed framework and the optimal value, respectively. This paper refers to the fraction of $J_{k\text{-GRAPE}}$ with respect to J_{OPT} as the suboptimality of $J_{k\text{-GRAPE}}$ denoted by α , i.e.,

$$\alpha := J_{k\text{-GRAPE}} / J_{\text{OPT}} \quad (14)$$

The lower bound of the suboptimality can be determined by the following theorem.

Theorem 4 (Suboptimality Lower Bound of k -GRAPE Algorithm): Given a Nash stable partition $\mathbf{P}$ obtained by Algorithm 1, its suboptimality in terms of the global utility is lower bounded as follows:

$$\alpha \geq J_{k\text{-GRAPE}} / (J_{\text{OPT}} + \lambda) \quad (15)$$

where

$$\lambda \equiv \sum_{\forall g_j \in \mathbf{P}} \max_{a_i \in \mathcal{A}, |g| \leq |\mathcal{A}|} \{ |g| \cdot [u_i(e_j, |g_j|, \mathbf{X}) - u_i(e_j, |g_j \cup \{a_i\}|, \mathbf{X})] \} \quad (16)$$

The proof of Theorem 4 is given in Appendix VI-E.

Although Theorem 4 does not provide a fixed-value lower bound, it can be determined as long as a Nash stable partition and EVUs' individual utility functions are given in any set of EVCS locations. Nevertheless, as a special case, if the social utility for any coalition is non-decreasing (or monotonically increasing) in terms of the number of co-visiting EVUs, then we can obtain a fixed-value lower bound for the suboptimality of a Nash stable partition.

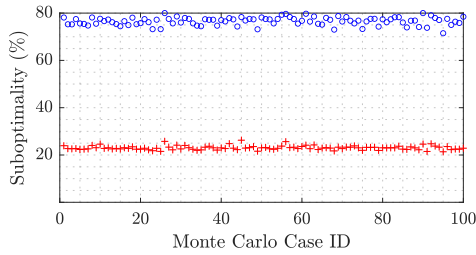


Fig. 5. True suboptimality of a Nash stable partition obtained by k -GRAPE for each instance (a blue circle) and its lower bound provided by Theorem 4 (a red cross).

Theorem 5 (Suboptimality Lower Bound of k -GRAPE Algorithm: a Special Case): Given an instance $(\mathcal{A}, \mathcal{E}, \mathcal{P})$ of Algorithm 1, if (i) the social utility for any group is non-decreasing with regard to the number of participants, i.e., for any $g_j \subseteq \mathcal{A}$ and $a_l \in \mathcal{A} \setminus g_j$, it holds that

$$\sum_{a_i \in g_j} u_i(e_j, |g_j|, \mathbf{X}) \leq \sum_{a_i \in g_j \cup \{a_l\}} u_i(e_j, |g_j \cup \{a_l\}|, \mathbf{X})$$

and (ii) if all the individual utilities can be derived from SPAO preference relations, then a Nash stable partition $\mathbf{P}$ obtained by Algorithm 1 provides at least 50% of suboptimality in terms of the global utility for any set of EVCS locations.

The proof of Theorem 5 is given in Appendix VI-F.

Monte Carlo simulations with 100 instances are performed to examine the suboptimality of the proposed framework. In each instance, there are $n_e = 3$ of EVCSs and $n_a = 12$ of EVUs in a given area (1000 km²). The EVCS brand utility function, Eq.(17), is applied. Figure 5 shows the true suboptimality of each instance ($J_{k\text{-GRAPE}}/J_{\text{OPT}}$), and the lower bound given by Theorem 4. In the figure, blue circles and red crosses indicate the true suboptimality and the lower bound, respectively. From the results, it can be seen that the proposed framework provides near-optimal solutions in most cases and the suboptimality of each Nash stable partition is enclosed by the corresponding lower bound.

k -GRAPE is the first attempt to model the charging station selection of EVUs using the anonymous hedonic game. In the proposed approach, each EVU receives the crowdedness of each EVCS and sends its decision to the EVCS. Therefore, based on the crowdedness information, each EVU selects an EVCS for its charging according to its individual preferences. In the following section, we present three experiments based on real-world-data from the point of view of EVUs.

V. NUMERICAL EXPERIMENTS

We run three experiments that determine locations of EVCSs based on individual strategies of heterogeneous EVUs, which have not been covered by the existing studies, on a real data set from Gothenburg. In *Case 1*, we show the effect of considering the crowdedness and travel distance with the individual brand preferences on user satisfaction and social benefits by comparing the Voronoi diagram approach, the weighted Voronoi diagram approach, and k -GRAPE. *Case 2*, we show the effect of considering the crowdedness and travel distance with the attraction preferences on user

satisfaction and social benefits by comparing the Voronoi diagram approach, the weighted Voronoi diagram approach, and k -GRAPE. In this case, additionally, we perform the sensitivity analysis with respect to an weighting parameter. In *Case 3*, the effect of the number of EVCSs is analyzed with regard to brand satisfaction through a parametric analysis. Then, we propose the number of EVCSs for each service provider (i.e., brand) to provide the best service to EVUs in terms of their preferred brand. Note that, the three simulations fulfill Assumption 1-4. We run the three case studies on MATLAB using a 2.81 GHz Intel processor with 16GB of RAM. We first describe the data and the baseline approach used for the experiments, and present the three case studies. Note that, due to that the existing approach based on the game theoretical framework in [7] only determines the number of EVCSs in the center of given districts, it is hard to quantitatively compare k -GRAPE with the existing game-based approach.

A. Data and Baseline Approach

The area considered for this experiments is shown in Figure 6 (89.10 km²). The maximum distance from east to west is 136.74 km and the maximum distance from north to south is 116.44 km. We utilize the population data of the area from Statistics Sweden for stationary charging cases such as charging during the weekend. According to the residence data, there are 361,995 people living in this area as of 2018. As there are hundreds of thousands of people, we made a basic assumption from [7], which is that the number of vehicles is proportional to the number of residents. Thus, it is assumed that 10% of the residents in Gothenburg have vehicles, and 5% of all the vehicles are EVs. The EVUs are black dots as shown in Figure 6. The capacity of each EVCS can indicate either the number of charging slots or the total electricity at the EVCS, indicating that the EVCSs can handle multiple EVs. We assume that every EVCS holds the same capacity in this study. Furthermore, we assume that the charging price per kilowatt-hour is the same in this study.

For the comparison, we utilize a weighted Voronoi diagram approach, which has been used in many positioning problems [42]–[44]. The Voronoi diagram approach is one of algorithms that place the EVCSs without geometric restriction, which is similar to the proposed approach. The iterative process of the Voronoi diagram approach is as follows: **(Step 1)** generating the corresponding Voronoi polygon of each EVCS based on the determined current position (initial position if it is the first iteration) of the EVCS, and computing the sum of the cost between EVUs and EVCSs; **(Step 2)** searching for the best position of the EVCSs that results in the local optimum cost in each Voronoi polygon and moving to the position; **(Step 3)** if it converges the iterative process is terminated, or else, back to **(Step 1)**.

B. Three Case Studies

1) *Case 1: Crowdedness and Travel Distance With Individual Brand Preferences:* In this case study, we show the effect of considering the crowdedness and travel distance with the individual brand preferences on user satisfaction and

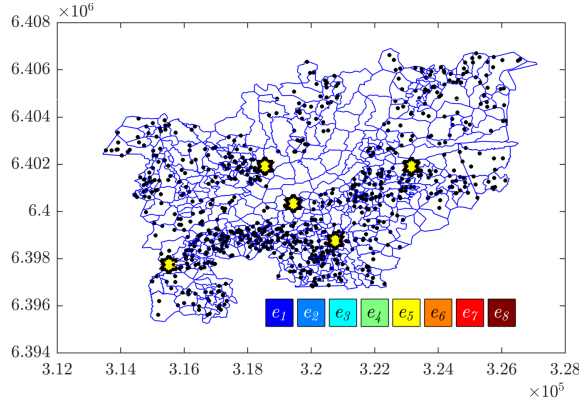


Fig. 6. Gothenburg center populations (each dot = 500 ppl), coordinate system SWEREF 99 TM. This shows visualized EVCS placement initial setting for *Case 1* and *Case 2* ($n_a = 752, n_e = 8, n_f = 5$). The dots indicate the positions of EVUs. The five yellow stars indicate the fixed position of the shopping malls in the city. (From the left: Orustgatan, Backaplan, Nordstan, Liseburg, and Torpavallsgatan). Each square label in right bottom represents each EVCS's position to be located and its size represents each of its capacity (same capacity in this case). [Source: Befolkning_1804_3006 ©SCB (Statistics Sweden)].

social benefits by comparing the Voronoi diagram approach, the weighted Voronoi diagram approach, and k -GRAPE. For k -GRAPE, the following utility function is considered individual:

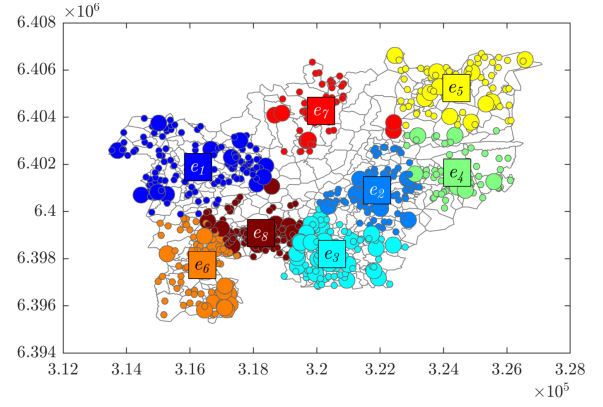
$$u_i(\mathbf{P}, \mathbf{X}) = u_i^{wd}(\mathbf{P}, \mathbf{X}) - w_{ij}^{brd} \cdot d_i(\mathbf{P}, \mathbf{X}), \quad (17)$$

where we categorize individual brand preference weights w_{ij}^{brd} split into six levels. The most preferred brand's weight is set to 1, the fifth most preferred brand's weight is set to 0.6, and all other weightings are set to 0.5. For example, for EVU a_i , the most preferred brand's weight w_{ij}^{brd} is 1, and the second most preferred brand's weight is 0.9. As we have no individual brand preferences data yet, we arbitrarily set up each user's brand preferences. For the weighted Voronoi diagram approach, we utilize the utility function Eq.(18), which applies the same weights of k -GRAPE.

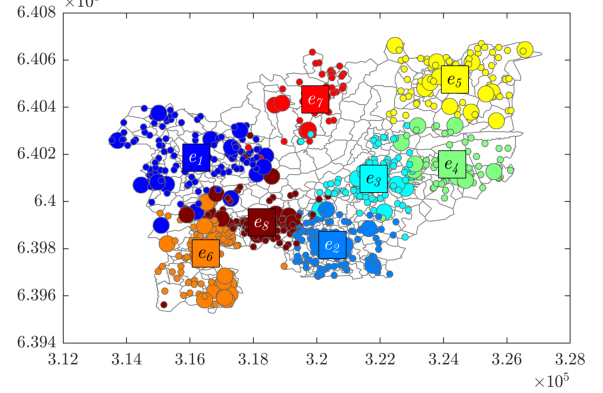
$$u_i(\mathbf{P}, \mathbf{X}) = -w_{ij}^{brd} \cdot d_i(\mathbf{P}, \mathbf{X}). \quad (18)$$

The same utility function is used in the Voronoi diagram approach but $w_{ij}^{brd} = 1, \forall a_i \in \mathcal{A}, \forall e_j \in \mathcal{E}$. Then, we solve the EVCSPP with eight EVCSs in Gothenburg.

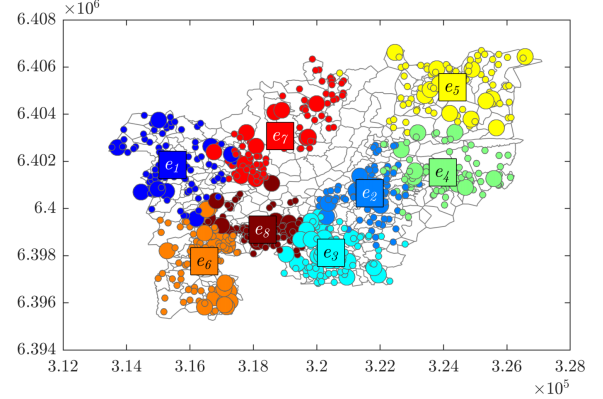
Figure 7 shows the results of the Voronoi diagram approach considering only the travel distance, those of the weighted Voronoi diagram approach considering the travel distance with the brand preference weights, and those of the proposed approach considering both crowdedness and the travel distance with brand preferences. We have stationary $n_a = 752$ EVUs and $n_t = 8$ EVCSs to be positioned. The initial locations of the EVCSs are randomly generated. The circles and squares indicate the positions of the EVUs and EVCSs, respectively. The size of the square indicates the capacity of each EVCS, which is assumed to be the same in this study. The colored EVUs are assigned to the same colored EVCS; for example, yellow colored EVUs are assigned to the yellow EVCS t_5 . The large circles indicate when the EVUs select the brand they prefer most.



(a) Voronoi diagram approach (without brand preferences) [16]



(b) Weighted Voronoi diagram approach (with brand preferences) [16]



(c) k -GRAPE (with brand preferences)

Fig. 7. Visualized EVCS placement results are shown with three approaches ($n_a = 752, n_e = 8$) in Gothenburg. Each square and its size represent each EVCS's position and its capacity, respectively. The circles indicate the position of EVUs. The color of each circle implies that the corresponding EVU's decision to the same colored EVCS (the large circles indicate that when the EVUs select the most preferred brand EVCS). Note that, the five attraction points are omitted in this case.

The result of the Voronoi diagram approach in Figure 7(a) shows the imbalance of the number of EVUs at the EVCSs. Particularly, significantly fewer EVUs are allocated to e_7 . In the case of the weighted Voronoi diagram approach in Figure 7(b), the individual brand preferences of EVUs are considered with the travel distance. EVUs with high preferences for a particular brand tend to select their preferred EVCS despite the existence of closer EVCSs. However, this approach has the disadvantage of an imbalance between supply and

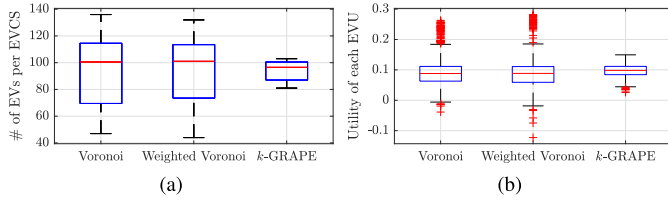


Fig. 8. Two box plots show two comparison results between three approaches: (a) the number of EVs per EVCS; (b) utility of each EVU. Note that, the results of the Voronoi diagram approach are same to the those of k -GRAPE without consideration of strategic behavior.

demand, which might cause congestion, an inefficient energy distribution and use of EVCSs. The result of the proposed approach in Figure 7(c) graphically shows a well-balanced result in terms of the number of EVs per EVCS by considering the crowdedness in addition to the individual brand preferences and the travel distance. Similar to Figure 7(b), it can be observed that some EVUs around the bounds between EVCSs tend to choose an EVCS with fewer EVUs and highly preferred brand, despite of the existence of closer EVCSs.

Figure 8(a) shows that the top 50% (median) of the EVCSs have 100.5 and 101 EVUs for the Voronoi diagram approach and the weighted Voronoi diagram approach, respectively, and 96.5 EVUs for k -GRAPE. For k -GRAPE, furthermore, the gap between the top 25% and remaining 75% is much smaller than with the baseline approaches. The supply and demand imbalance between the EVCSs is one of the drawbacks of the two baseline approaches. The results of the utility in Figure 8(b) show that the median utility of k -GRAPE is 10.47% and 10.12% larger than those in the Voronoi diagram approach and the weighted Voronoi diagram approach, respectively. Also, the figure shows that the two baseline approaches have more outliers compared with the result of the proposed approach. The outliers indicate the EVUs having more utility than the median. In other words, the two baseline approaches yield an improper distribution of resources between the EVUs.

With regard to the sum of travel distances of all EVUs to their corresponding EVCS, the result of k -GRAPE is 5.81% and 2.94% larger than those in the Voronoi diagram approach and the weighted Voronoi diagram approach, respectively. At the expense of such degradation (social costs) in the total travel distance, k -GRAPE can provide results satisfactory to the EVUs with regard to utility function and the balanced capacity distribution (social benefits). In other words, the proposed approach, which considers the crowdedness and travel distance with the individual preferences of EVUs, can distribute the given resources in an efficient and equitable way.

In order to capture the effect of the crowdedness, we perform the sensitivity analysis with respect to an increment of the crowdedness weighting parameter. The following utility function based on Eq.(17) is considered:

$$u_i(\mathbf{P}, \mathbf{X}) = w_{cud}^{sen} \cdot u_i^{cud}(\mathbf{P}, \mathbf{X}) - w_{brd}^{sen} \cdot w_{ij}^{brd} \cdot d_i(\mathbf{P}, \mathbf{X}), \quad (19)$$

where $w_{cud}^{sen} + w_{brd}^{sen} = 1$, and w_{cud}^{sen} and w_{brd}^{sen} are crowdedness and brand weighting for the sensitivity analysis, respectively. As we can expect from the weighting effect, Figure 9 shows

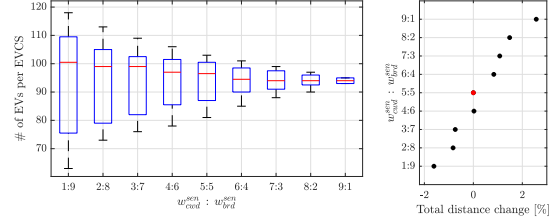


Fig. 9. Sensitivity analysis with regard to the weighting parameters: the number of EVs per EVCS (left); total distance change, i.e., social cost (right). Note that the red dot on the right figure is considered as a reference value.

that as the crowdedness weighting increases the distribution of EVUs becomes more even, although total distance increases.

The range anxiety is a factor affecting the driving behavior of EVUs. In order to discuss the uncertainties with regard to the battery capacity and weather, which are directly related to the range of EV, we conduct a Monte-Carlo simulation with a conservative problem setting (20% lower battery capacity, low departure state of charge of the EVs). The results showed that the uncertainties have little impact on the EVCSPP due to the average travel range of EVs (2.57 km^2) and the size of the considered area (89.10 km^2). Therefore, the uncertainties are expected to play a more important role in a large geographical area.

2) *Case 2: Crowdedness and Travel Distance With Individual Attraction Preferences*: To capture the effect of local attractions on the behavior of EVUs, in this case, we compare the EVCSPP with local attractions with the EVCSPP without local attractions. As we focus on the effect of the attraction preferences of EVUs on their EVCS selection, we assume that the attractions in this study are in the same service category (i.g., all the attractions are shopping centers). The service category can be easily changed by weighting different values. Furthermore, we assume that the EVUs have a more preferred attraction among them, and the closest shopping mall from each EVU is the most preferred attraction. The rest of the simulation settings for this case are the same as in *Case 1*, but we include five shopping malls to investigate their effect. As shown in Figure 6, there are five shopping malls and the location of eight EVCSs ($e_1, e_2, \dots, e_8$) should be selected considering the preferences of EVUs with regard to the local attractions.

Utility Function: A possible scenario is that the EVUs might often charge their battery on the way to a target destination, rather than only set out for charging. Thus, we assume that the EVUs would prefer to visit EVCSs located near attractions. For the case with attractions, we consider multi-criteria utility functions by combining the crowdedness, travel distance and the local attraction utility function, Eq.(4), which can be described as follows:

$$u_i(\mathbf{P}, \mathbf{X}) = w_{ijk}^{att} \cdot u_i^{cud}(\mathbf{P}, \mathbf{X}) - d_i(\mathbf{P}, \mathbf{X}), \quad (20)$$

where the attraction weight w_{ijk}^{att} is denoted for EVU a_i with regard to EVCS e_j and attraction f_k in which we only consider the closest attraction f_k from EVCS e_j . For the case without attraction, we utilize the following utility function:

$$u_i(\mathbf{P}, \mathbf{X}) = u_i^{cud}(\mathbf{P}, \mathbf{X}) - d_i(\mathbf{P}, \mathbf{X}). \quad (21)$$

TABLE II

SELECTION PERCENTAGE OF EVCSs NEAR EVUs' PREFERRED SHOPPING MALL: (A) k -GRAPE WITHOUT CONSIDERATION OF THE ATTRACTIONS; (B) WEIGHTED VORONOI DIAGRAM APPROACH WITH CONSIDERATION OF THE ATTRACTION; (C) k -GRAPE WITH CONSIDERATION OF THE ATTRACTIONS

	(a)	(b)	(c)
1 st pref. attraction [%]	86.44	91.09	94.81
2 nd pref. attraction [%]	9.84	6.65	4.39
3 rd pref. attraction [%]	2.39	1.73	0.80
4 th pref. attraction [%]	1.33	0.53	0
5 th pref. attraction [%]	0	0	0

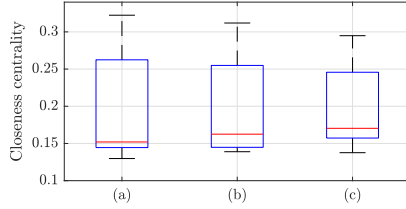


Fig. 10. The plot shows each case in Figure 6 changes normalized closeness centrality of the attractions. (a) k -GRAPE without consideration of local attractions; (b) Weighted Voronoi diagram approach with consideration of local attractions; (c) k -GRAPE with consideration of local attractions.

We simply assign the same weight w_{ijk}^{att} in Eq.(20) to the weighted Voronoi diagram approach in reverse order, and the weight is multiplied by the travel distance $d_i(\mathbf{P}, \mathbf{X})$. This utility function indicates that an EVCS near a highly preferred attraction has lower costs to EVUs.

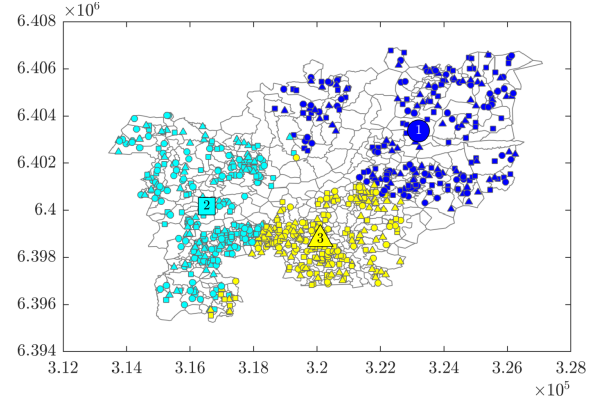
Table II shows the selection percentage of charging stations in view of the preferred shopping malls of EVUs for three cases. For the case of k -GRAPE, approximately 95 % of EVUs select the EVCS near their most preferred shopping mall. The other two cases show relatively lower satisfaction than the k -GRAPE case.

As the preferences are determined by the travel distance, we can deduce that the EVCSs are located near the shopping malls. We utilize a centrality measure to analyze the effect of the attractions quantitatively. The *closeness centrality* measures the reciprocal of the sum of the length of the shortest paths between the shopping malls and all other EVUs via an EVCS within the region [45]. The normalized closeness centrality used in this study is as follows:

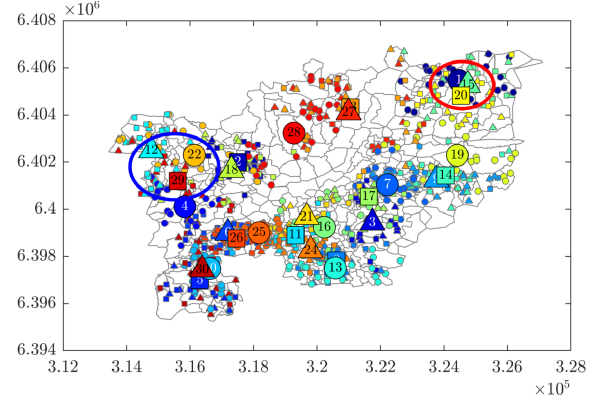
$$C_c(f_k) = \frac{1}{\sum_{a_i \in \mathcal{A}} d_{ik} + \sum_{t_j \in \mathcal{T}} d_{jk}}. \quad (22)$$

The median values in Figure 10 show that the importance of the shopping malls of the case (b) is larger than in the case (a), and the corresponding importance in case (c) is greater than that in the other two cases. That is, this comparison quantitatively shows that the EVCSs congregate in the shopping malls. In practice, more EVUs can easily access the shopping malls. Consequently, service providers should note that considering attractions increases the accessibility and satisfaction of EVUs.

3) *Case 3: Parametric Analysis With Respect to the Number of EVCSs*: Different service providers should facilitate the distribution of the appropriate electrical power in response to the demand. We perform a parametric analysis by changing the number of EVCSs using utility function Eq.(17). This case study can be applied to determine the number of public



(a) 3 EVCSs placement result (3 service providers install 1 EVCS each)



(b) 30 EVCSs placement result (3 service providers install 10 EVCSs each)

Fig. 11. EVCS placement results are shown with the different number of the EVCSs ($n_e = 3, 30, n_a = 752$). Each large circle, triangle, and square, and its size represent each EVCS's brand (and position) and its capacity (same capacity in this case), respectively. Each small circle, triangle, and square indicate each EVU, and its most preferred brand. The color of each EVU indicates that the corresponding their decision to the same colored EVCS: (a) 3 EVCSs placement result; (b) 30 EVCSs placement result.

charging stations instead of optimizing the number of EVCSs in the districts [7]. Here, it is assumed that there are 3 EVCS brands (Circle A:○, Square B:□, Triangle C:△), and each of the 3 service providers should install their EVCSs in Gothenburg. We perform simulations by increasing the number of EVCSs by 3 from 3 to 36 (i.e., each service provider adds one EVCS). The other simulation settings are the same as in Case 2.

Figure 11 shows two visualized simulation results of the cases with 3 EVCSs and 30 EVCSs. The results show a trend whereby the three brands congregate at the places where relatively many EVUs exist; for example, EVCS 1 (brand A:○), EVCS 20 (brand B:□), and EVCS 15 (brand C:△) in the red circle at the right top of Figure 11(b), and EVCS 22 (brand A:○), EVCS 29 (brand B:□), and EVCS 12 (brand C:△) in the blue circle at the left top of Figure 11(b). Thus, we can observe that, regardless of the other brands, each service provider will install their EVCSs in a high-demand area. Namely, the crowdedness mainly influences the decisions of EVUs. Figure 12(a) shows that, as the number of EVCSs increases the percentage of EVUs that select their preferred brand increases in the area. The percentage of selection of the

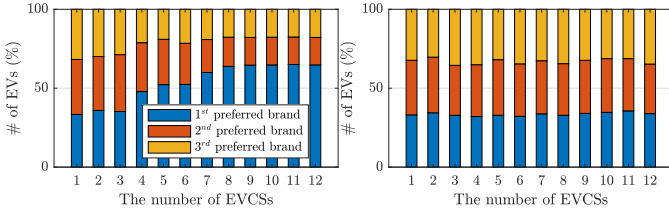


Fig. 12. The plot shows the brand preferences effect: (a) with the brand preferences effect; (b) without the brand preferences effect.

most preferred charging station converges to 64.8 %, which might because of the components of the utility function in Eq(17). The trade-off between the crowdedness and the travel distance with individual brand preferences causes to converge to the value. In other words, 35.2 % of users select less crowded 2nd or 3rd preferred EVCSs because even if the EVUs consider their brand preferences, they still value on the crowdedness. Therefore, when service providers plan to install public charging stations in Gothenburg, it is recommended to place at least nine EVCSs to attract as many customers who have brand preferences as possible when there are other service providers.

VI. CONCLUSION

Reasonable placement of EVCSs is an important premise for the rapid development of the EV industry. The proposed algorithm, *k*-GRAPE, aims to attract investments to the electric car industry by determining the position of the fast-charging stations effectively based on the individual preferences of each EVU. Our contributions are as follows: 1) *k*-GRAPE allows strategic and competitive decision-making of the EVUs depending on crowdedness and their preferences with privacy preserved data from the EVCSs; 2) it is proven that the proposed algorithm converges to a local optimum in a finite number of iterations and the solution at least guarantees 50% of suboptimality; 3) *k*-GRAPE proposes geographically unconstrained individual EVCS locations that satisfy the preferences of EVUs; 4) numerical experiments using the real-world data from Gothenburg in Sweden show that *k*-GRAPE outperforms the baseline approach in terms of balanced energy distribution, satisfaction of EVUs and closeness centrality.

Here, some weaknesses of the proposed approach are pointed out for further studies. First, we only have shown that the suboptimality of the proposed algorithm is 50%, even if the numerical results provide near-optimal solutions. Second, although this framework can consider individual preferences and crowdedness from the customer perspective, it cannot include the service providers' viewpoint. One natural progression of this study is to consider both the EVU's perspective and the fast-EVCSs' perspective. Finally, due to the limitation of the data access, we assume the individual preferences of EVUs. One interesting study would be to consider real-world data for modeling the EVUs. This will improve the fidelity of the EVU model in this field and practicalize this framework.

Further research might explore a cooperative way to solve the EVCSPP, i.e., it is assumed that customers cooperate with each other to maximize their utilities. This could be expected in public transportation cases such as car-sharing services.

APPENDIX ANALYSIS

A. Proof of Lemma 1

Proof: For (1), given $\mathbf{X}$ and $\mathbf{P}$, for EVU a_i , the number of additional EVUs tolerable in its group and it is defined:

$$\Delta_{\mathbf{P}(i)} := \min_{g_j \in \mathbf{P} \setminus \{g_{\mathbf{P}(i)}\}} \max_{\Delta \in \mathbb{Z}^+} \{\Delta | u_i(\mathbf{P}', \mathbf{X}) \geq u_i(\mathbf{P}, \mathbf{X})\},$$

where the number of EVUs who select t_j given $\mathbf{X}$ and $\mathbf{P}$ is $|g_j \cup \{a_i\}|$, and the number of EVUs who select $t_{\mathbf{P}(i)}$ given $\mathbf{X}$ and $\mathbf{P}'$ is $|g_{\mathbf{P}(i)} + \Delta|$. Due to the SPAO condition, this value satisfies the following characteristics: (a) if $\mathbf{P}$ is Nash stable, for every EVU a_i they holds $\Delta_{\mathbf{P}(i)} \geq 0$; (b) $\Delta_{\mathbf{P}(i)} < 0$ indicates that EVU a_i will deviate to another group in the next iteration; and (c) for the EVU a_i who deviated at the last iteration and updated the partition as $\mathbf{P}'$ holding that $\Delta_{\mathbf{P}(i)} \geq 0$.

For (2), from Definition 1, it is clear that the new instance $(\hat{\mathcal{A}}, \mathcal{E}, \hat{\mathcal{P}})$ holds SPAO. Let $\mathbf{P}_0$ denote a Nash stable partition in $(\mathcal{A}, \mathcal{E}, \mathcal{P})$. When a new EVU $a_r \notin \mathcal{A}$ decides to charge at one of the EVCSs in $\mathcal{E}$ and creates a new partition $\mathbf{P}_1$, it holds that $\Delta_{\mathbf{P}(r)} \geq 0$. If no EVU $a_q \in \mathcal{A}$ whose $\Delta_{\mathbf{P}(q)} < 0$, then the new partition $\mathbf{P}_1$ is Nash stable.

For (3), suppose that there is at least an EVU a_q whose $\Delta_{\mathbf{P}(q)} < 0$. The EVU must be one of the members in the group that EVU a_r selected in the last iteration. As EVU a_q moves to another group and creates a new partition $\mathbf{P}_2$, the previous deviated user a_r holds $\Delta_{\mathbf{P}_1(q)} \geq 1$. I.e., an EVU who deviates to a group and changes one of the current users in the group will not deviate again. It indicates that at most $|\hat{\mathcal{A}}|$ of iterations are required to hold $\Delta_{\hat{\mathbf{P}}(q)} \geq 0$ for every EVU $a_i \in \hat{\mathcal{A}}$, where the partition $\hat{\mathbf{P}}$ is Nash stable. $\square$

B. Proof of Theorem 1

Proof: This theorem will be proved by induction. Let $T(n)$ be the following mathematical statement: for $|\mathcal{A}| = n$, if an instance $(\mathcal{A}, \mathcal{E}, \mathcal{P})$ is SPAO, then there exists a Nash stable partition.

Base Case: When $n = 1$, only an EVU exists in an instance $(\mathcal{A}, \mathcal{E}, \mathcal{P})$. The user is able to participate in its most preferred group and the resultant partition is Nash stable. Therefore, $T(1)$

Induction Hypothesis: Assume that $T(k + 1)$ is true for a positive integer such that $|\mathcal{A}| = k$

Inductive Step: Suppose that a new EVU $a_i \notin \mathcal{A}$ whose preferences relation regarding every EVCS in $\mathcal{E}$ is SPAO participates the instance $(\mathcal{A}, \mathcal{E}, \mathcal{P})$. This induces a new instance $(\hat{\mathcal{A}}, \mathcal{E}, \hat{\mathcal{P}})$ where $\hat{\mathcal{A}} = \mathcal{A} \cup \{a_i\}$ and $|\mathcal{A}| = k + 1$. From Definition 1, it is clear that the new instance also satisfies SPAO and has a Nash stable. Let $\mathbf{P}_0$ denote a Nash stable partition in the original instance. When a new EVU $a_i \notin \mathcal{A}$ decides to select one of the EVCSs in $\mathcal{E}$ and creates a new partition $\mathbf{P}_1$, it holds that $\Delta_{\mathbf{P}_1(i)} \geq 0$ (i.e., EVU a_i deviated at the last iteration and updated the partition as $\mathbf{P}_1$). If there are no more EVUs $a_r \in \mathcal{A}$ whose $\Delta_{\mathbf{P}_1(r)} < 0$, then the new partition $\mathbf{P}_1$ is Nash stable. Consequently, $T(k + 1)$ is true. By mathematical induction, $T(k)$ is true for all positive integers $k \geq 1$. $\square$

C. Proof of Theorem 2

Proof: Suppose that, given a Nash stable partition in an instance where there is only one EVU, another arbitrary EVU is added in the instance. Then, we find a Nash stable partition for this new instance, and repeat the procedure until all the EVUs in $\mathcal{A}$ are included. From Lemma 1, if a new EVU joins an instance in which the current partition is Nash stable, then the maximum number of iterations required to find a new Nash stable partition is the number of the existing EVUs plus one. Therefore, it is trivial that the maximum number of iterations to find a Nash stable partition of an instance $(\mathcal{A}, \mathcal{E}, \mathcal{P})$ is given as

$$\sum_{k=1}^{|\mathcal{A}|} k = |\mathcal{A}| \cdot (|\mathcal{A}| + 1)/2. \quad (23)$$

□

D. Proof of Theorem 3

Proof: Let $\Phi(\mathbf{P}^{(t)}, \mathbf{X}^{(t)}) = \sum_{\forall a_i \in \mathcal{A}} u_i(\mathbf{P}^{(t)}, \mathbf{X}^{(t)})$, where $\mathbf{P}^{(t)}$ is a set of n_e current partition $\{g_1^{(t)}, g_2^{(t)}, \dots, g_{n_e}^{(t)}\}$ and $\mathbf{X}^{(t)}$ is a set of n_e current EVCS locations $\{x_1^{(t)}, x_2^{(t)}, \dots, x_{n_e}^{(t)}\}$. Suppose that Algorithm 3 proceeds from iteration t to iteration $t + 1$, then it is sufficient to show the followings: we first show that given $\mathbf{P}$ Algorithm 2 always finds exact solution $\mathbf{X}^*$ within a finite number of iterations; and, then we show $\Phi(\mathbf{P}^{(t+1)}, \mathbf{X}^{(t)}) \leq \Phi(\mathbf{P}^{(t+1)}, \mathbf{X}^{(t+1)})$ (improvement).

The first step shows that Algorithm 1 finds the same $\mathbf{X}^*$ for the same partition $\mathbf{P}^t$ regardless of iteration t . For example, if two partition $\mathbf{P}^t$ and $\mathbf{P}^{t'}$ are the same, its following EVCS location update results are the same (i.e., $\mathbf{X}^t = \mathbf{X}^{t'}$). Therefore, this step makes the algorithm recursively iterates within a finite number of iteration.

The second step can be proven by showing that for any fixed EVCS locations, the sum squared error is minimized when the location associated with each EVCS is the centroid of the group assigned to the EVCS. Here, as partition $\mathbf{P}^{t+1}$ is the same for both sides (i.e., only the Euclidean distance changes the sum of utility function), we can consider $\Phi(\mathbf{P}^{(t+1)}, \mathbf{X}^{(t+1)})$ as the sum squared error defined as $\text{SSE}(\mathbf{P}^{(t+1)}, \mathbf{X}^{(t+1)}) = \sum_{j=1}^{n_e} \|a_{\forall g_j}^i - \mathbf{X}_{\mathbf{P}^{(t+1)}}^{(t+1)}(j)\|^2$. Then, we can show $\Phi(\mathbf{P}^{(t+1)}, \mathbf{X}^{(t)}) \leq \Phi(\mathbf{P}^{(t+1)}, \mathbf{X}^{(t+1)})$:

$$\begin{aligned} \text{SSE}(\mathbf{P}^{(t+1)}, \mathbf{X}^{(t)}) &= \sum_{j=1}^{n_e} \|a_{\forall g_j}^i - \mathbf{X}_{\mathbf{P}^{(t+1)}}^{(t)}(j)\|^2 \\ &\leq \sum_{j=1}^{n_e} \|a_{\forall g_j}^i - \mathbf{X}_{\mathbf{P}^{(t+1)}}^{(t+1)}(j)\|^2 \\ &= \text{SSE}(\mathbf{P}^{(t+1)}, \mathbf{X}^{(t+1)}). \end{aligned}$$

Due to Algorithm 1 only guarantee a certain level of optimality, we cannot guarantee that a solution of Algorithm 3 monotonically increase. Based on the property of the first step, we can find $\mathbf{P}^*, \mathbf{X}^*$ that maximizes $u_i(\mathbf{P}^{t'}, \mathbf{X}^{t'})$ in $\mathcal{D}'$. □

E. Proof of Theorem 4

Proof: Let $\mathbf{P}^*$ denote the optimal partition for the objective function in Eq.(14). Given a Nash stable partition $\mathbf{P}$, from

Definition 3, it holds that $\forall a_i \in \mathcal{A}$

$$u_i(e_{\mathbf{P}(i)}, |g_{\mathbf{P}(i)}|, \mathbf{X}) \geq u_i(e_{j \leftarrow i}^*, |g_j \cup \{a_i\}|, \mathbf{X}) \quad (24)$$

where $e_{j \leftarrow i}^*$ indicates EVCS $e_j \in \mathcal{E}$ to which EVU a_i should have joined according to the optimal partition $\mathbf{P}^*$; and $g_j \in \mathbf{P}$ is the group for EVCS e_j whose participants follow the Nash stable partition $\mathbf{P}$. The right-hand side of Eq.(24) can be rewritten as

$$\begin{aligned} &u_i(e_{j \leftarrow i}^*, |g_j \cup \{a_i\}|, \mathbf{X}) \\ &= u_i(e_{j \leftarrow i}^*, |g_j^*|, \mathbf{X}) \\ &\quad - \{u_i(e_{j \leftarrow i}^*, |g_j^*|, \mathbf{X}) - u_i(e_{j \leftarrow i}^*, |g_j \cup \{a_i\}|, \mathbf{X})\} \end{aligned} \quad (25)$$

where $g_j^* \in \mathbf{P}^*$ is the ideal coalition of task $e_{j \leftarrow i}^*$ that maximizes the objective function.

By assuming over all the agents, Eq.(24) can be expressed as

$$\begin{aligned} &\sum_{\forall a_i \in \mathcal{A}} u_i(e_{\mathbf{P}(i)}, |g_{\mathbf{P}(i)}|, \mathbf{X}) \\ &\leq \sum_{\forall a_i \in \mathcal{A}} u_i(e_{j \leftarrow i}^*, |g_j^*|, \mathbf{X}) \\ &\quad - \sum_{\forall a_i \in \mathcal{A}} \{u_i(e_{j \leftarrow i}^*, |g_j^*|, \mathbf{X}) - u_i(e_{j \leftarrow i}^*, |g_j \cup \{a_i\}|, \mathbf{X})\}. \end{aligned} \quad (26)$$

The left-hand side of Eq.(26) represents the objective function value of the Nash stable partition $\mathbf{P}$, i.e., $J_{k\text{-GRAPE}}$, and the first term of the right-hand side is the optimal value, i.e., J_{OPT} . The second term in the right-hand side can be interpreted as the summation of the utility loss of each EVU caused by the belated decision to its optimal EVCS, provided that the other EVUs still follow the Nash stable partition.

The upper bound of the second term is given by

$$\sum_{j=1}^{n_e} |g_j^*| \cdot \max_{a_i \in g_j^*} \{u_i(e_{j \leftarrow i}^*, |g_j^*|, \mathbf{X}) - u_i(e_{j \leftarrow i}^*, |g_j \cup \{a_i\}|, \mathbf{X})\}. \quad (27)$$

This is at most

$$\sum_{\forall g_j \in \mathbf{P}} \max_{a_i \in \mathcal{A}, |g| \leq |\mathcal{A}|} L_{ij}[g] \equiv \lambda \quad (28)$$

where $L_{ij}[g] = |g| \cdot (u_i(e_j, |g|, \mathbf{X}) - u_i(e_j, |g_j \cup \{a_i\}|, \mathbf{X}))$.

Hence, Eq.(26) can be rewritten as

$$J_{k\text{-GRAPE}} \geq J_{\text{OPT}} - \lambda$$

Dividing both sides by $J_{k\text{-GRAPE}}$ and rearranging them yield the suboptimality lower bound of the Nash stable partition as given by Eq.(15) □

F. Proof of Theorem 5

Proof: First, we introduce some definitions and notations that facilitate to describe this proof. Given a partition $\mathbf{P}$ of an instance $(\mathcal{A}, \mathcal{E}, \mathcal{P})$, the global utility is denoted by

$$V(\mathbf{P}) := \sum_{\forall a_i \in \mathcal{A}} u_i(e_{\mathbf{P}(i)}, |g_{\mathbf{P}(i)}|, \mathbf{X}). \quad (29)$$

We use operator $\oplus$ as follows. Given any two partitions $\mathbf{P}^A = \{g_0^A, \dots, g_{n_e}^A\}$ and $\mathbf{P}^B = \{g_0^B, \dots, g_{n_e}^B\}$

$$\mathbf{P}^A \oplus \mathbf{P}^B := \{g_0^A \cup g_0^B, g_0^A \cup g_1^B, \dots, g_{n_e}^A \cup g_{n_e}^B\}.$$

Since $\cup_{j=0}^{n_e} g_j^A = \cup_{j=0}^{n_e} g_j^B = \mathcal{A}$, there may exist the same EVU a_i even in two different group in $\mathbf{P}^A \oplus \mathbf{P}^B$. For instance, suppose that $\mathbf{P}^A = \{\{a_1\}, \{a_2\}, \{a_3\}\}$ and $\mathbf{P}^B = \{\{\emptyset\}, \{a_1, a_3\}, \{a_2\}\}$. Then, $\mathbf{P}^A \oplus \mathbf{P}^B = \{\{a_1\}, \{a_1, a_2, a_3\}, \{a_2, a_3\}\}$. We regard such an EVU as two different agents in $\mathbf{P}^A \oplus \mathbf{P}^B$. Accordingly, the operation may increase the number of total EVUs in the resultant partition.

Using the definitions described above, condition (i) implies that 14

$$V(\mathbf{P}^A) \leq V(\mathbf{P}^A \oplus \mathbf{P}^B) \quad (30)$$

From now on, we will show that $\frac{1}{2}V(\mathbf{P}^*) \leq V(\hat{\mathbf{P}})$, where $\mathbf{P}^* = \{g_0^*, g_1^*, \dots, g_{n_e}^*\}$ is an optimal partition and $\hat{\mathbf{P}} = \{\hat{g}_0, \hat{g}_1, \dots, \hat{g}_{n_e}\}$ is a Nash stable partition. By doing so, this theorem can be proved. From the definition in Eq.(29), it can be said that

$$V(\hat{\mathbf{P}} \oplus \mathbf{P}^*) = \sum_{\forall a_i \in \mathcal{A}} u_i(e_{\hat{\mathbf{P}}(i)}, |\hat{g}_{\hat{\mathbf{P}}(i)} \cup g_{\mathbf{P}^*(i)}^*|, \mathbf{X}) + \sum_{\forall a_i \in \mathcal{A}^-} u_i(e_{\mathbf{P}^*(i)}, |\hat{g}_{\hat{\mathbf{P}}(i)} \cup g_{\mathbf{P}^*(i)}^*|, \mathbf{X}) \quad (31)$$

where $\mathcal{A}^-$ is the set of EVUs whose decisions follow not the Nash stable partition $\hat{\mathbf{P}}$ but only the optimal partition $\mathbf{P}^*$. Due to condition (ii), the first term of the right-hand side in Eq.(31) is no more than

$$\sum_{\forall a_i \in \mathcal{A}} u_i(e_{\hat{\mathbf{P}}(i)}, |\hat{g}_{\hat{\mathbf{P}}(i)}|, \mathbf{X}) \equiv V(\hat{\mathbf{P}}). \quad (32)$$

Likewise, the second term is also at most

$$\sum_{\forall a_i \in \mathcal{A}^-} u_i(e_{\mathbf{P}^*(i)}, |\hat{g}_{\hat{\mathbf{P}}(i)} \cup \{a_i\}|, \mathbf{X}). \quad (33)$$

By the definition of Nash stability (i.e., for every EVU $a_i \in \mathcal{A}$, $u_i(e_{\hat{\mathbf{P}}(i)}, |\hat{g}_{\hat{\mathbf{P}}(i)}|, \mathbf{X}) \geq u_i(e_j, |\hat{g}_j \cup \{a_i\}|, \mathbf{X})$, $\forall \hat{g}_j \in \hat{\mathbf{P}}$), the above equations is at most

$$\sum_{\forall a_i \in \mathcal{A}^-} u_i(g_{\hat{\mathbf{P}}(i)}, |\hat{g}_{\hat{\mathbf{P}}(i)}|, \mathbf{X}) \quad (34)$$

which is also no more than, because of $\mathcal{A}^- \subseteq \mathcal{A}$

$$\sum_{\forall a_i \in \mathcal{A}} u_i(e_{\hat{\mathbf{P}}(i)}, |\hat{g}_{\hat{\mathbf{P}}(i)}|, \mathbf{X}) \equiv V(\hat{\mathbf{P}}). \quad (35)$$

Accordingly, the left-hand side of Eq.(31) holds the following inequality:

$$V(\hat{\mathbf{P}} \oplus \mathbf{P}^*) \leq 2V(\hat{\mathbf{P}}) \quad (36)$$

Thanks to Eq.(30), it follows that

$$V(\mathbf{P}^*) \leq V(\hat{\mathbf{P}} \oplus \mathbf{P}^*) \quad (37)$$

Therefore, $V(\mathbf{P}^*) \leq 2V(\hat{\mathbf{P}})$ for any EVCS placement $\mathbf{X}$, which completes this proof. $\square$

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scheduling algorithm of

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